

Given that ecdysteroid can cause either molting or metamorphosis, what determines which process takes place? The answer is another signal, juvenile hormone (JH), secreted by a pair of endocrine glands behind the brain. JH modulates ecdysteroid activity. When the level of JH in body fluids is high, ecdysteroid stimulates molting (and thus maintains the “juvenile” larval state). When the JH level drops, ecdysteroid instead induces formation of a pupa, within which metamorphosis occurs.

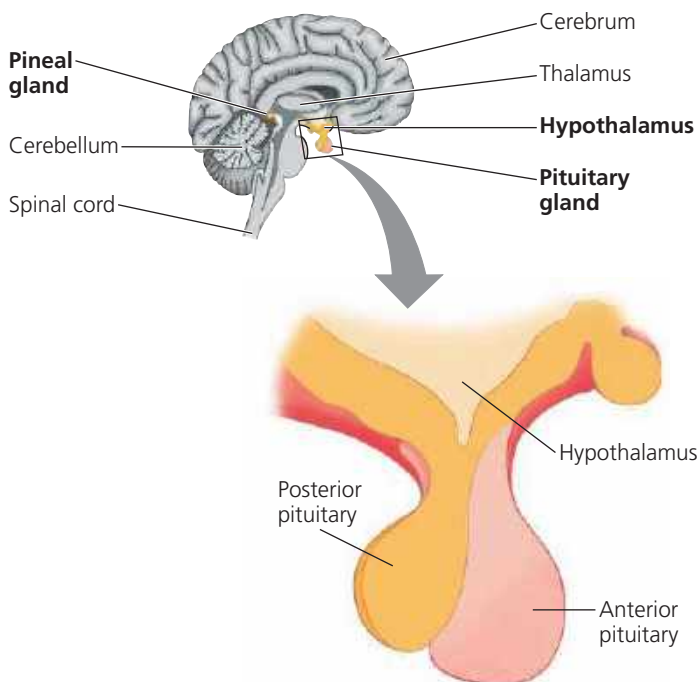
Knowledge of the coordination between the nervous system and endocrine system in insects has provided a basis for novel methods of agricultural pest control. For example, one tool to control insect pests is a chemical that binds to the ecdysteroid receptor, causing insect larvae to molt prematurely and die.

Vertebrates

In vertebrates, coordination of endocrine signaling relies heavily on the **hypothalamus** (Figure 45.13). The hypothalamus receives information from nerves throughout the body and, in response, initiates neuroendocrine signaling appropriate to environmental conditions. In many vertebrates, for example, nerve signals from the brain pass sensory information to the hypothalamus about seasonal changes. The hypothalamus, in turn, regulates the release of reproductive hormones required during the breeding season.

Signals from the hypothalamus travel to the **pituitary gland**, a gland located at the base of the hypothalamus

▼ **Figure 45.13 Endocrine glands in the human brain.** This side view of the brain indicates the position of the hypothalamus, the pituitary gland, and the pineal gland. (The pineal gland plays a role in regulating biological rhythms.)



(see Figure 45.13). Roughly the size and shape of a lima bean, the pituitary is made up of two glands that fused during development but remain as discrete posterior and anterior parts, or lobes, that perform very different functions. The **posterior pituitary** is an extension of the neural tissue of the hypothalamus. Hypothalamic axons that reach into the posterior pituitary secrete neurohormones synthesized in the hypothalamus. In contrast, the **anterior pituitary** is an endocrine gland that synthesizes and secretes hormones in response to hormones from the hypothalamus.

Posterior Pituitary Hormones Neurosecretory cells of the hypothalamus synthesize the two posterior pituitary hormones: antidiuretic hormone (ADH) and oxytocin. After traveling to the posterior pituitary within the long axons of the neurosecretory cells, these neurohormones are stored, to be released into the bloodstream in response to nerve impulses transmitted by the hypothalamus (Figure 45.14).

Antidiuretic hormone (ADH), or *vasopressin*, regulates kidney function. Circulating ADH increases water retention in the kidneys, helping maintain normal blood osmolarity (see Concept 44.5). ADH also has an important role in social behavior (see Concept 51.4).

Oxytocin has multiple functions related to reproduction. As we have seen, in female mammals oxytocin controls milk secretion by the mammary glands and regulates uterine contractions during birthing. In addition, oxytocin has targets in

▼ **Figure 45.14 Production and release of posterior pituitary hormones.** The posterior pituitary gland is an extension of the hypothalamus. Certain neurosecretory cells in the hypothalamus make antidiuretic hormone (ADH) and oxytocin, which are transported to the posterior pituitary, where they are stored. Nerve signals from the brain trigger release of these neurohormones.

